		stone must remain the same.
13	**B**	Using $T^2 \propto R^3$, $(\frac{27.3}{T})^2 = (\frac{R}{R/30})^3 \Rightarrow R = 0.166 \text{ day } \approx 4.0 \text{ hour}$
14	**A**	Conservation of energy, $E_{Final} = E_{Initial}$